Build Guide for Tree Imaging Machine

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1 Installing Jetson Linux 36.3.0 and Other Software Dependencies

1.1 Download Drivers and Follow NVIDIA Quick Start Guide

Jetson Linux 36.3.0 must first be installed on the Jetson Orin Nano Developer Kit using an x86-based computer running Ubuntu 22.04 or Ubuntu 20.04.

1. Download the *Driver Package (BSP)* and *Sample Root Filesystem* for Jetson Linux 36.3.0 onto an x86 Ubuntu 22.04 or Ubuntu 20.04 computer.
2. Follow the NVIDIA Quick Start Guide. Ensure that the environment variables for the Driver Package and Root Filesystem point to the correct file paths on your computer.
3. Connect the Jetson to a computer monitor, keyboard, and mouse.
4. Complete the first-boot setup. Once the Jetson has booted, confirm the installed Jetson Linux version by running the following command in the terminal:

```
cat /etc/nv_tegra_release
```

The terminal should display output similar to the following, indicating that Jetson Linux Version 36.3 has been installed:

```
# R36 (release), REVISION:3.0, ...
```

1.2 Connect the Raspberry Pi HQ Camera to the Jetson

The Raspberry Pi HQ Camera is now ready to be connected to the Jetson.

1. Unplug the Jetson from any power source.
2. Take out the 15 to 22 pin flexible camera cable.
3. Pull on the plastic pull tab on the CAM1 connector of the Jetson to open the connector.
4. Insert the narrower, 22 pin end of the cable into the connector, making sure the metal contacts are facing the circuit board, NOT the sky. Push in the plastic tab to secure the cable.
5. Pull on the plastic pull tab on the connector on the Raspberry Pi HQ Camera to open the connector.
6. Insert the wider, 15 pin end of the cable into the camera connector, making sure the metal contacts are facing the circuit board, NOT the sky. Push in the plastic tab to secure the cable.

1.3 Downloading the GitHub Repository

Once Jetson Linux has been installed, the TIM GitHub repository must be downloaded onto the Jetson.

1. Log into the Jetson and connect to the internet using Wi-Fi or Ethernet.
2. Clone the TIM GitHub repository using Git:

```
git clone https://github.com/ANONYMIZED/REPO
cd REPO
```

1.4 Installing Dependencies Using Docker

The TIM system is designed to be ran inside a Docker container for ease of installation and portability. If you enter any command and receive a permission error, run the same command with `sudo` at the beginning and enter your password when prompted.

1. Make the build, launch, and entrypoint scripts executable:

```
chmod +x build.sh launch.sh entrypoint.sh
```

2. Run the build script to install the required dependencies. This process may take several minutes:

```
./build.sh
```

3. The launch script will start a new Docker container with the necessary dependencies. Once the container launches, the terminal environment will change to the Docker container.

```
./launch.sh
```

4. Start the Python GUI by running the following command inside the Docker container. This will launch two windows: one displaying the live camera viewer and another displaying the user interface.

```
python gui.py
```

1.5 Checking if the Docker Container is Running

Another way to confirm that the Docker container is running, open a new terminal tab without closing the terminal currently running the `./launch.sh` process. Run the following command in the new terminal:

```
sudo docker ps
```

This will display a list of currently active Docker containers. Confirm that the project container is listed.

2 Assembling the ACRO Motion Kit

2.1 OpenBuilds ACRO 1010 Kit + OpenBuilds Linear Actuator Kit + OpenBuilds BlackBox Motor Controller

We based our design on the OpenBuilds ecosystem, relying on their motion kits and motor controller. You must follow their instructions for the initial assembly steps. They are detailed below.

1. Assemble the ACRO by following the assembly video made by OpenBuilds (OpenBuilds, 2019a).
2. Assemble the linear actuator by following the assembly video made by OpenBuilds (OpenBuilds, 2018). The linear actuator will eventually need to be attached to the X and Y planes, as described in later steps.
3. Follow the electrical assembly instruction video made by OpenBuilds (OpenBuilds, 2019b).

After following these detailed videos, you will have a complete motion system connected to the BlackBox. The controller is ready to receive G-code commands to guide its motion. Testing motion is described in Section 3.

2.2 Attach the OpenBuilds V-Slot NEMA 17 Linear Actuator to the ACRO

The ACRO needs the linear actuator to be attached with custom adapters to include motion in the Z direction.

2.3 Alternative Motion Kit Suggestions

Motion kits are subject to market availability. There are many kits available, and the most suitable options are sold as CNC router kits, which can be readily adapted to hold a camera and Jetson Orin Nano. So long as the motor controller has GRBL firmware, it will be compatible with the commands that TIM uses to control the motion. Assembly instructions vary by kit. Adapters must be designed and 3D printed to fit any kit other than the OpenBuilds ACRO. Please write a GitHub Issue if you don't know where to start in order to design a 3D-printed bracket for your machine kit.

Product Name	Manufacturer	City	Country	Price (CAD)	Product Link
4040-PRO Desktop CNC Router (400 × 400 × 78 mm)	Genmitsu	Lenexa	USA	$752	Genmitsu
FoxAlien Masuter Pro 3-Axis CNC Router (400 × 400 × 60 mm)	FoxAlien	Dongguan	China	$670	Amazon Canada
VEVOR CNC Router 300W, 3-Axis (400 × 400 × 75 mm)	VEVOR	Shanghai	China	$550	Amazon Canada
Optional Extension Kits					
4040-PRO Working Area Extension Kits	Genmitsu	Lenexa	USA	$200 to $400	Genmitsu

Table 1: Alternative CNC motion kits and available extension options.

3 Testing XYZ Motion with Universal G-Code Sender Platform

https://github.com/winder/Universal-G-Code-Sender/wiki/Usage#ugs-platform

4 Install Adapters, Jetson, and Camera to the ACRO

4.1 Attaching Leg Extensions

Add additional height to the ACRO using aluminum extrusion. Use the corner brackets to connect the aluminum extrusions and form the legs as seen in FIgure 1. The 3D-printed leg height tool makes it easy to ensure that all four legs are the same length. The necessary components are listed in Table 2.

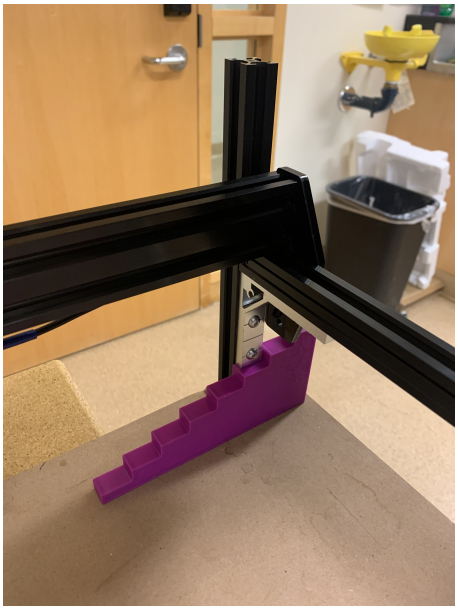


Figure 1: 3D-printed leg height tool used to ensure consistent leg lengths.

Component Type	Component Name	Quantity
3D Prints	leg_height_tool.stl	1
Hardware	M4 × 12 mm bolts	16
	M4 hammer head t nuts	16
Extra	Extruded aluminum corner brackets	4
	250 mm × 20 mm × 20 mm aluminum extrusion	4

Table 2: Components required for attaching the ACRO legs.

4.2 Attaching the Camera and Computer Adapter

Bolt on the camera and computer adapters to the linear actuator kit using bolts and locknuts. The exploded view of the assembly is shown in Figure 2. The necessary components are listed in Table 3.

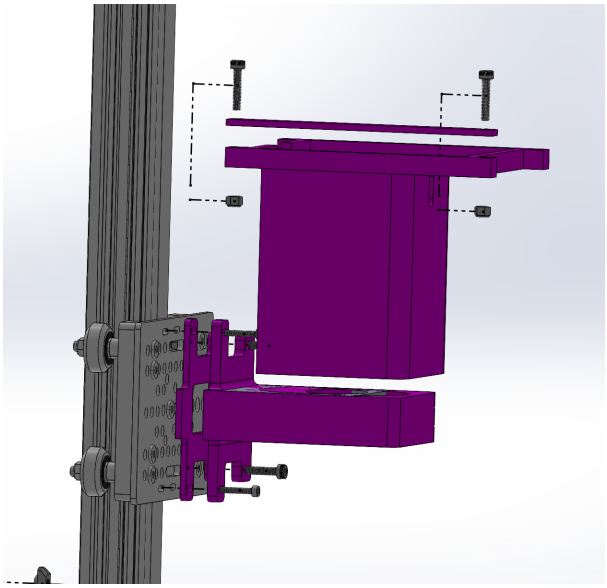


Figure 2: Exploded view of the camera and computer adapter assembly.

Component Type	Component Name	Quantity
3D Prints	z_axis_plate_adapter.stl	1
	z_axis_plate_spacer.stl	1
	z_axis_jetson_carriage.stl	1
	z_axis_jetson_lock.stl	1
Hardware	M3 × 16 mm bolts	2
	M3 locknuts	2
	M4 × 16 mm bolts	2
	M4 locknuts	2
Extra	OpenBuilds Linear Actuator attachment plate	1

Table 3: Components required for the camera and computer adapter.

4.2.1 Attaching the Z Axis

The linear actuator needs to be connected to the ACRO gantry to allow for motion in the Z direction. Use the hammer head t-nuts to mount into the aluminum extrusion of the linear actuator. The exploded view of the assembly is shown in Figure 3. The necessary components are listed in Table 4.

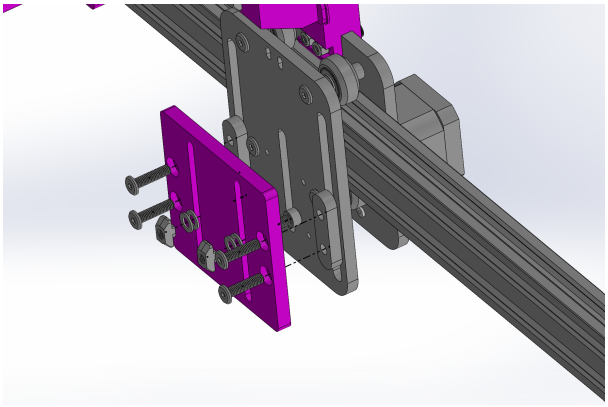


Figure 3: Exploded view of the Z-axis adapter assembly.

Component Type	Component Name	Quantity
3D Prints	z_axis_adapter.stl	1
	lens_plate_spacer.stl	1
Hardware	M5 × 16 mm bolts	2
	M5 T-nuts	2
	M5 washers	6
Extra	OpenBuilds Linear Actuator	1

Table 4: Components required for attaching the Z axis.

The attachment plate supplied with the ACRO is not designed to easily connect to the linear actuator. To address this, we designed a custom Z-axis adapter with slots that align with the aluminum extrusion.

The washers are used as spacers to shim between the bolts and the adapter, allowing adjustments to the effective bolt length and ensuring a secure fit.

4.3 Torsion Correction

The additional weight of the computer, camera, lens, and linear actuator results in some torsional deflection of the ACRO gantry. It is important for the camera to be parallel to the surface on which the samples are placed to ensure consistent image acquisition. To compensate for deflection in the gantry beam, the torsion correction system allows shims to be added or removed. These shims adjust the angle of the linear actuator relative to the imaging table. You can rotate the Z-axis by removing or adding shims as necessary. Use a bullet level to verify that the camera assembly is level relative to the imaging surface. The exploded view of the torsion correction assembly is shown in Figure 4. The necessary components are listed in Table 5.

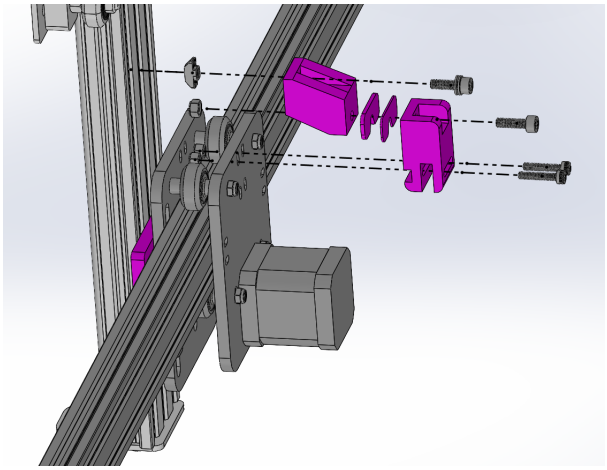


Figure 4: Exploded view of the torsion correction assembly.

Component Type	Component Name	Quantity
3D Prints	torsion_correction_plate_adapter.stl	1
	torsion_correction_extrusion_adapter.stl	1
	torsion_2mm_shim.stl	2
	torsion_1mm_shim.stl	2
Hardware	M5 × 16 mm bolts	2
	M5 T-nuts	1
	M5 washers	2
	M5 locknut	1
	M4 × 20 mm bolts	2
	M4 locknuts	2
Extra	OpenBuilds Linear Actuator	1

Table 5: Components required for the torsion correction system.

4.4 Installing the Camera and Computer

Now that the Z axis has been added to the ACRO, install the camera with the lens attached into its designated mounting position underneath the Jetson Orin Nano. Place the Jetson Orin Nano into the carriage and tighten the locking mechanism to prevent it from falling during operation. Route the wires alongside the cables shown in the ACRO system electronics video (OpenBuilds, 2019b).

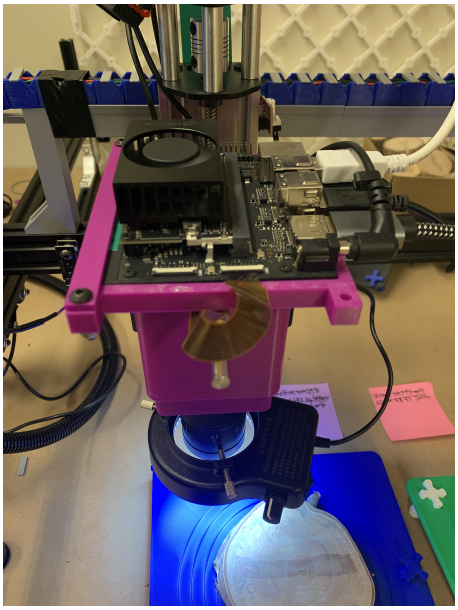


Figure 5: Jetson Orin Nano installed in the carriage with the camera assembly mounted underneath.

Component Type	Component Name	Quantity
Extra	Jetson Orin Nano	1
	Raspberry Pi HQ Camera	1
	SEEED Studio Lens	1
	CSI cable	1

Table 6: Components required for camera and computer installation.

5 Operating TIM

6 References

[1] OpenBuilds (2019a). OpenBuilds Acro System Assembly (Mechanical Build) [Video]. YouTube. https://www.youtube.com/watch?v=YDLJrxrMEAo

[2] OpenBuilds (2018). V-Slot® NEMA 17 Linear Actuator Bundle (Lead Screw) [Video]. YouTube. https://www.youtube.com/watch?v=X0Z_gfl-iGY

[3] OpenBuilds (2019b). OpenBuilds Acro System Electronics and Software (Using BlackBox) [Video]. YouTube. https://www.youtube.com/watch?v=GrjqW2MDCvM